

Lab: Descriptive statistics

Estimated reading time: 30 minutes

In this lab, you'll learn to configure Data Analysis ToolPak. You'll also be able to explore how to use Descriptive Statistics in Excel. You'll also be able to interpret the results of descriptive statistical analysis.

Software Used in this Lab

The instruction videos in this course use the full Excel Desktop version, as this has all the available product features. However, for the hands-on labs, we will use the free 'Excel for the web' version as this is available to everyone.

Although you can use the Excel Desktop software if you have access to this version, [it is recommended that you use Excel for the web for the hands-on labs](#), as the lab instructions specifically refer to this version. There are some small differences in the interface and available features.

Data Set Used in this Lab

The data set used in this lab is curated from the following source: <https://www.kaggle.com/datasets/yasserh/housing-prices-dataset> under the [CC0: Public Domain license](#).

Acknowledgment and thanks also go to <https://trak.in>, who were generous enough to share the data publicly for free.

We are using a modified subset of that data set for the lab. To follow the lab instructions successfully, please use the data set provided in this lab rather than the data set from the original source.

Learning Objectives:

- Configure Data Analysis ToolPak
- Use the Descriptive Statistics function in Excel to analyze a data set
- Interpret results of descriptive statistical analysis

Welcome to the lab: Descriptive Statistics!

Exercise: Descriptive Statistics

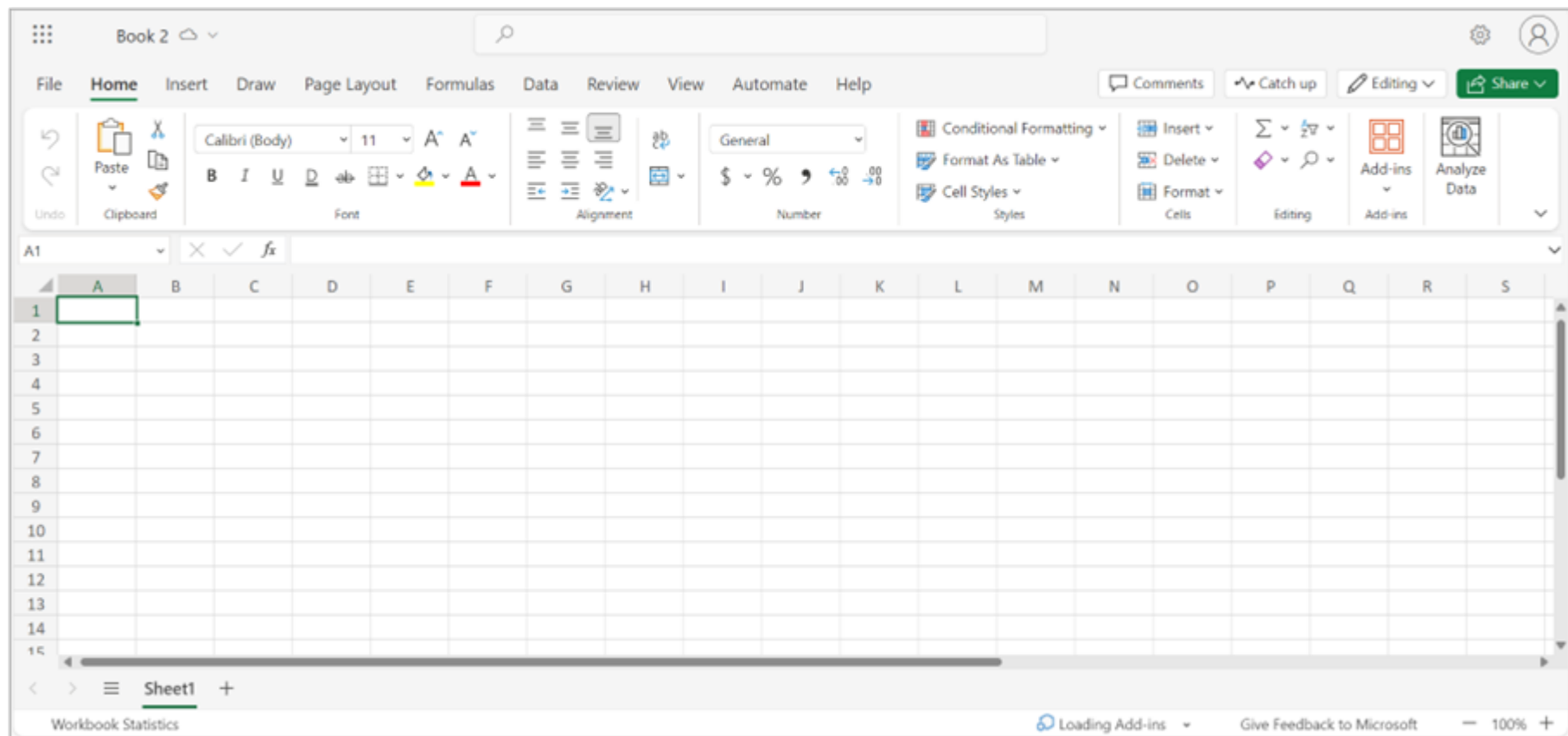
In this exercise, you'll configure the add-in XLMiner Analysis ToolPak available in the Excel for the web. You'll also measure central tendency, dispersion, skewness, and kurtosis using Descriptive statistics.

Note: XLMiner Analysis ToolPak is a third-party free statistical analysis ToolPak in Excel for the web. It shows the same functions as the Analysis ToolPak add-in available in Excel for the desktop. You can use this ToolPak to perform Descriptive Statistics and Linear Regression for the labs in this course. If you are using Excel for the desktop, you can configure in-built Data Analysis ToolPak available from Microsoft.

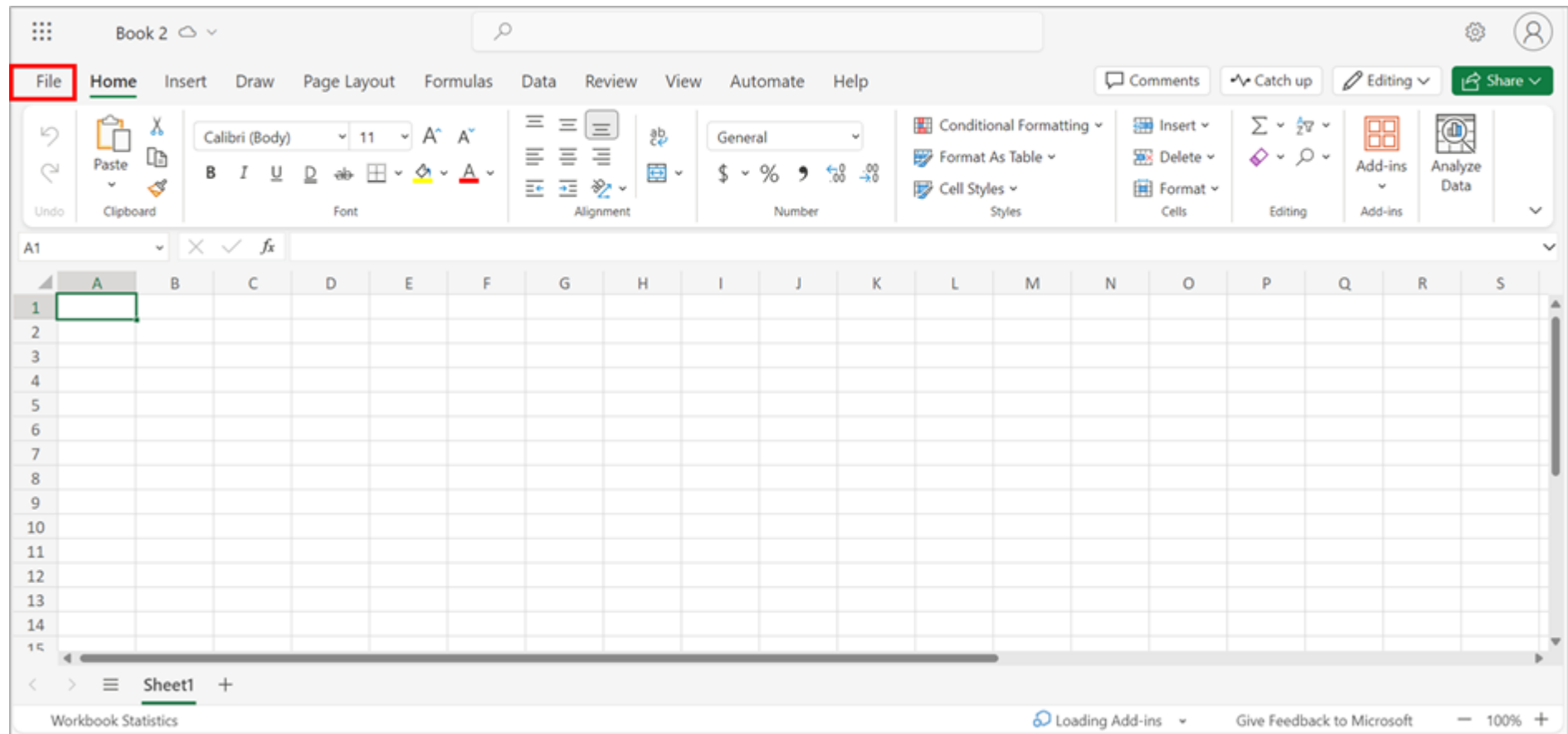
Task 1: Configure XLMiner Analysis ToolPak

In this task, you'll configure XLMiner Analysis ToolPak in Excel for the web. To do so, log in to Excel for the web and perform the steps below.

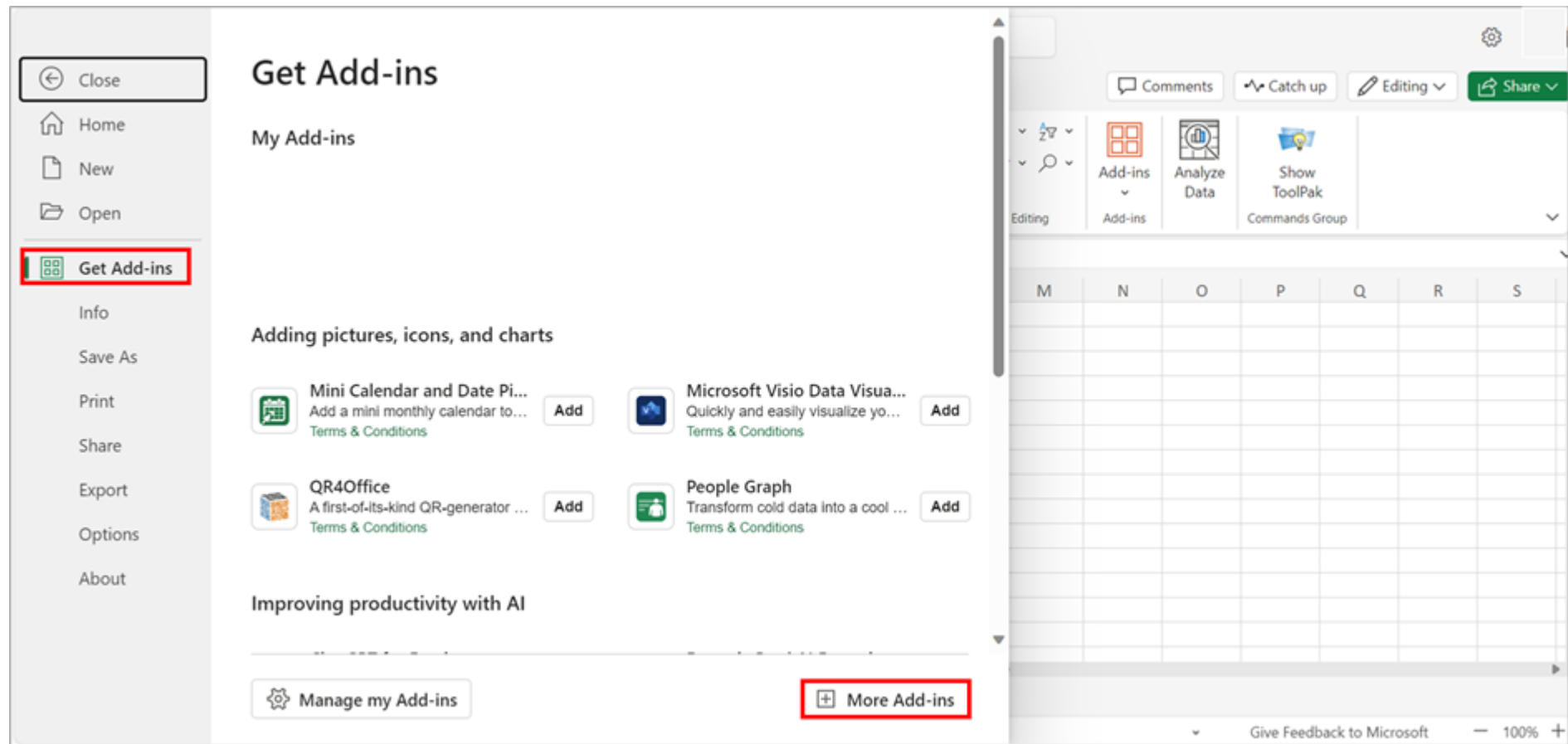
1. Open Excel for the web on your computer or laptop.



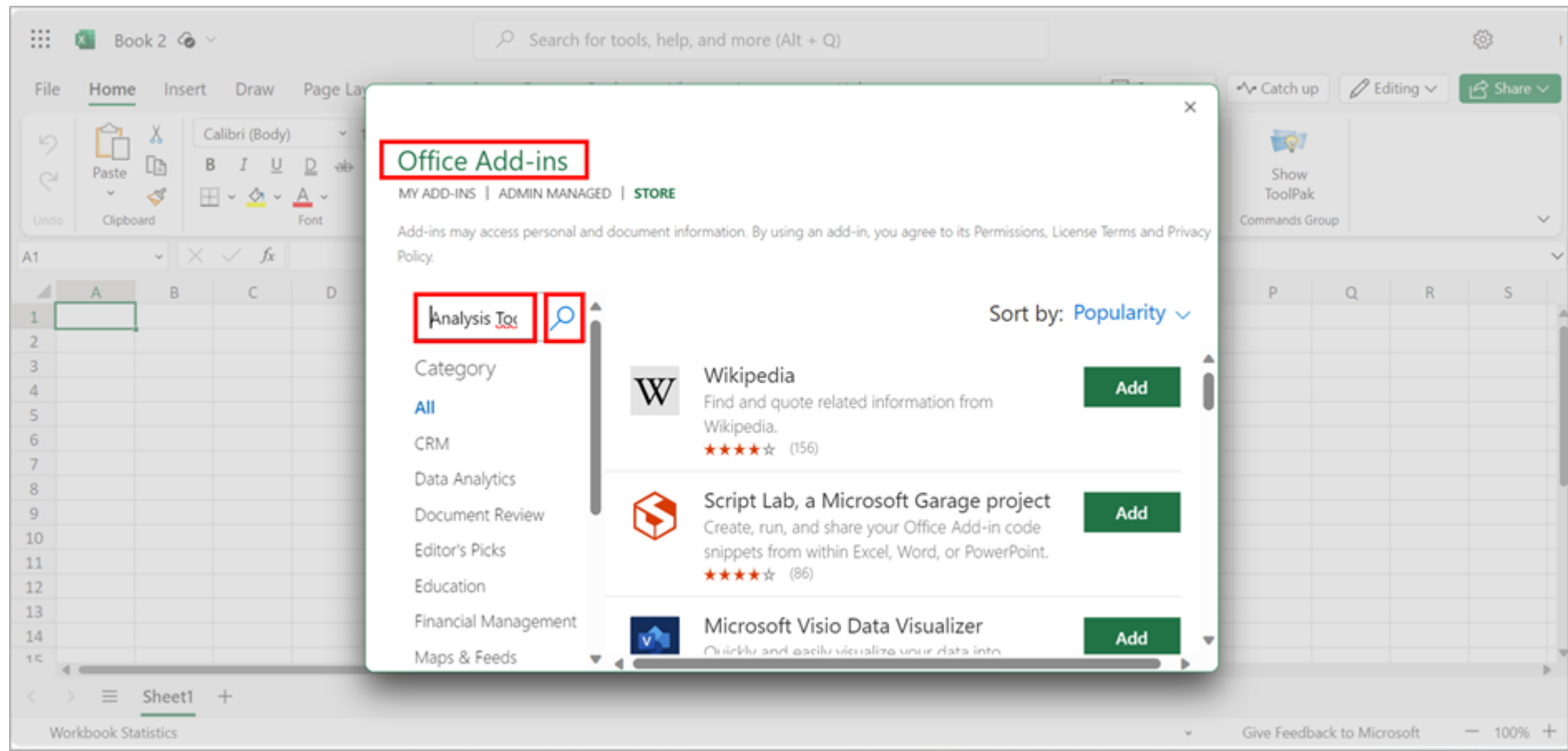
2. Select **File** on the top ribbon.



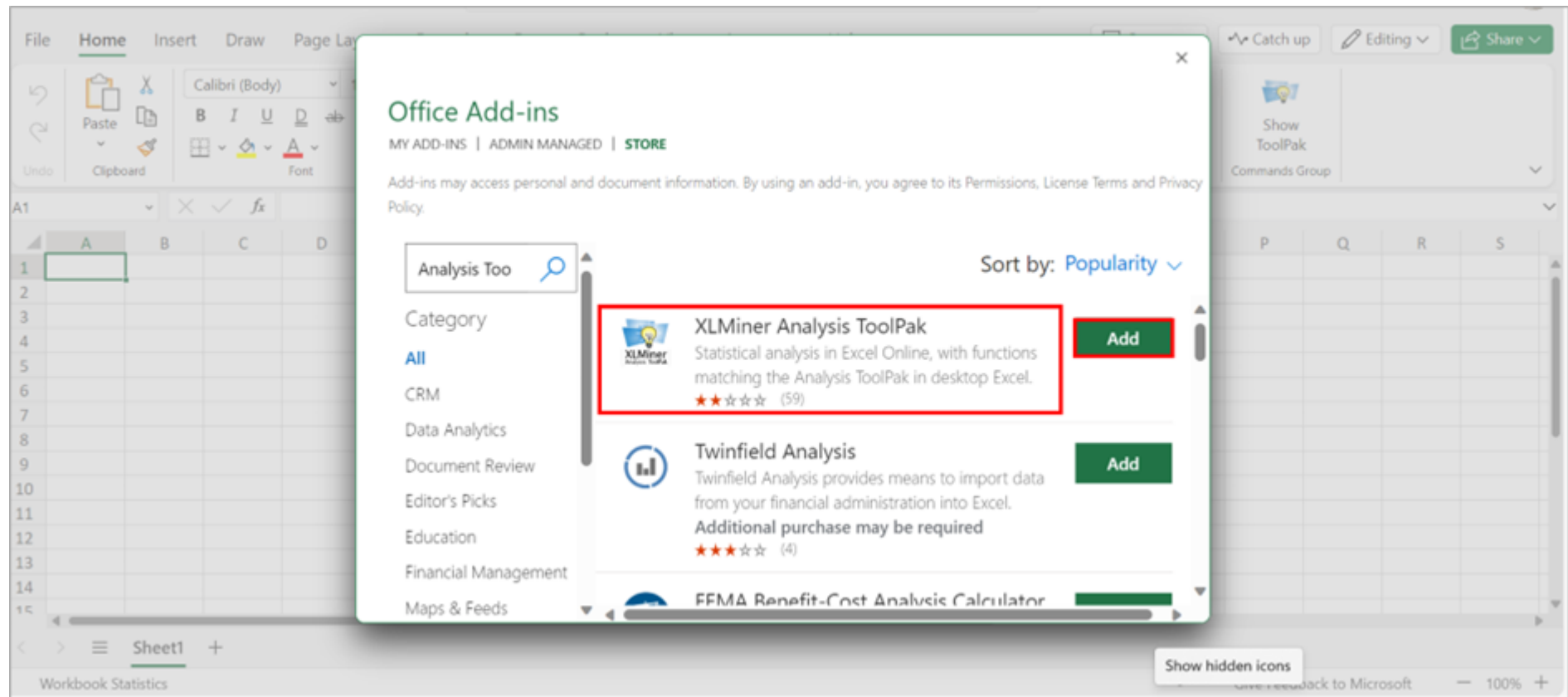
3. Select **Get Add-ins** from the left menu bar and choose **More Add-ins**.



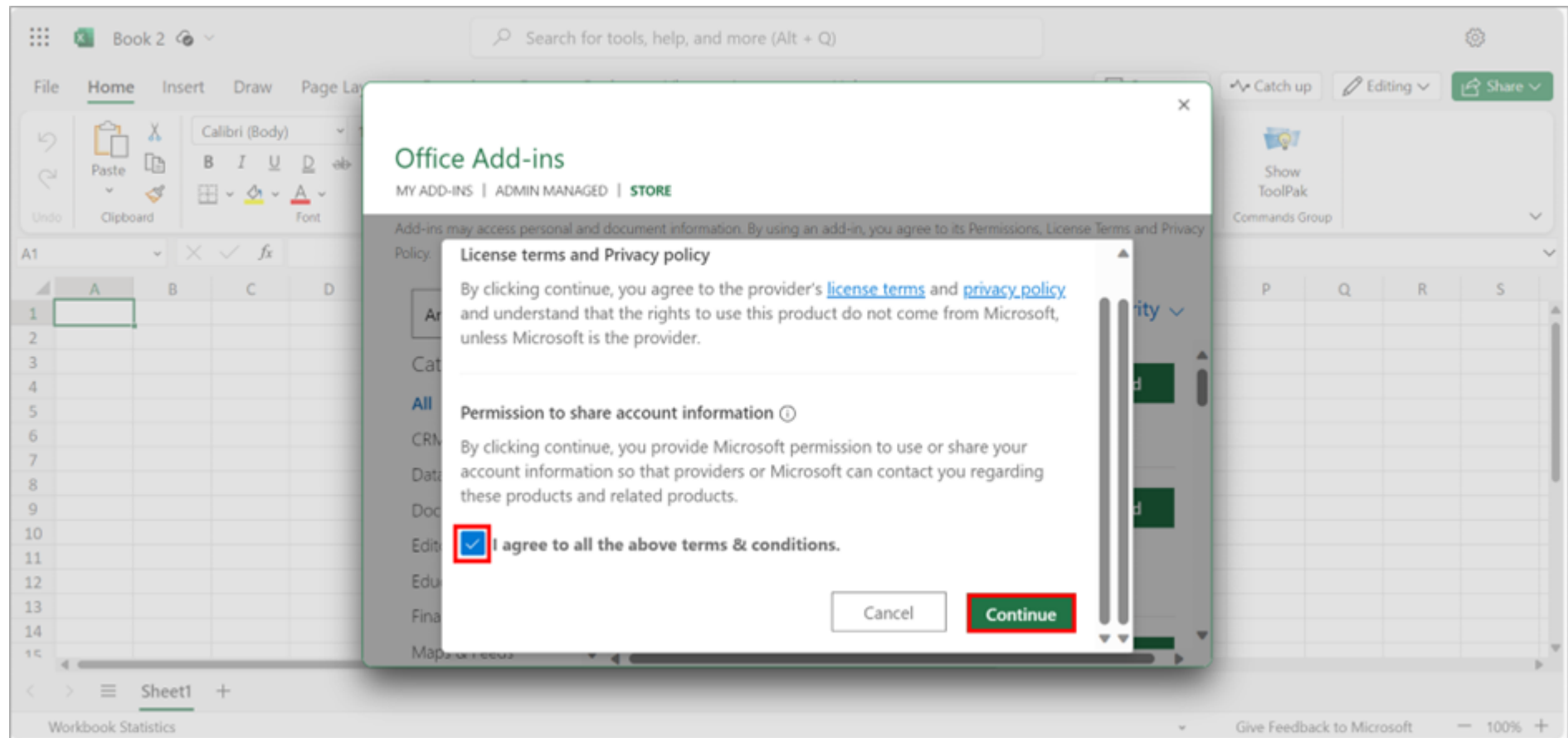
4. A window **Office Add-ins** will appear. In the **Search** section, type in the keyword **Analysis ToolPak** and select the search icon.



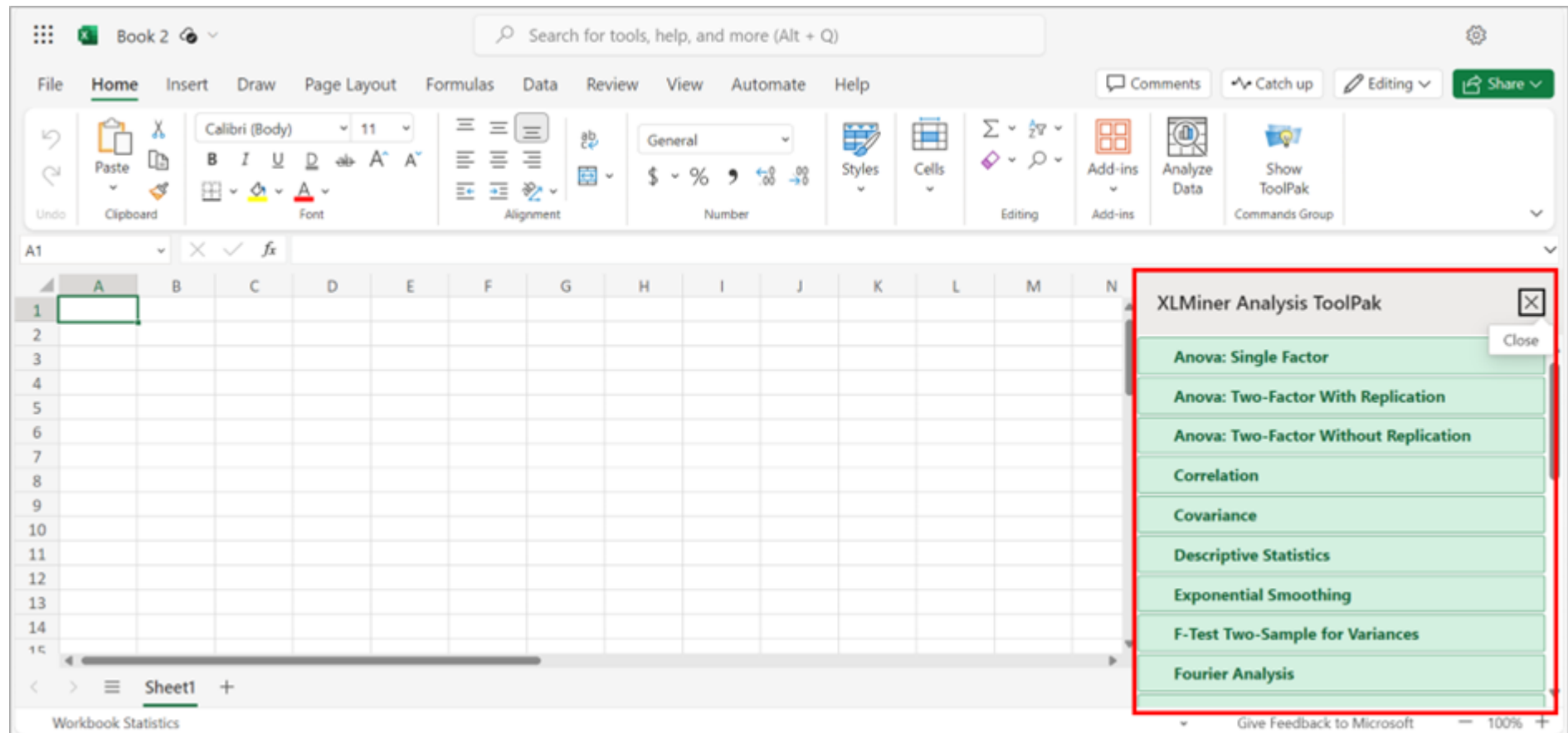
5. An **XLMiner Analysis ToolPak** will appear. Select **Add**.



6. Checkmark **I agree to all the above terms & conditions** for license terms and privacy policy. Select **Continue**.



7. You'll see **XLMiner Analysis ToolPak** on the right side of your Excel sheet.



Task 2: Calculate Descriptive Statistics Using Excel

In this task, you'll learn to calculate a measure of central tendency and dispersion using Descriptive Statistics using Excel. You'll also be able to calculate and interpret skewness and kurtosis. Follow the below steps to perform these calculations.

1. Copy and paste the data set from the source data into an Excel worksheet. In this case, you can use the data set for the [housing prices](#) (To download the given data set in this lab, right click on the given hyperlink and select Open link in new tab).

The screenshot displays the Microsoft Excel interface. The 'Home' tab is active on the ribbon. A dataset is entered in the range A1:D11, which is highlighted with a red border. The data includes house numbers, areas, interest rates, and prices. The 'XLMiner Analysis ToolPak' task pane is open on the right side of the window, showing a list of statistical analysis options.

House	Area (sq. ft.)	Bank's Interest Rate (%)	Housing Price (USD)
1	1500	3	450,000
2	2000	3.5	520,000
3	1200	2.5	340,000
4	1800	3.2	500,000
5	1600	2.8	430,000
6	1400	2.2	480,000
7	1300	2.6	320,000
8	1700	3.3	480,000
9	2100	3.1	560,000
10	2300	3.4	600,000

XLMiner Analysis ToolPak

- Anova: Single Factor
- Anova: Two-Factor With Replication
- Anova: Two-Factor Without Replication
- Correlation
- Covariance
- Descriptive Statistics
- Exponential Smoothing
- F-Test Two-Sample for Variances
- Fourier Analysis

2. Select **Descriptive Statistics** from the **XLMiner Analysis ToolPak**.

The screenshot displays the Microsoft Excel interface with the XLMiner Analysis ToolPak add-in installed. The ribbon is set to 'Home'. The spreadsheet contains data for 10 houses, with columns for House number, Area (sq. ft.), Bank's Interest Rate (%), and Housing Price (USD). The XLMiner Analysis ToolPak menu is open on the right side, and the 'Descriptive Statistics' option is highlighted with a red box.

House	Area (sq. ft.)	Bank's Interest Rate (%)	Housing Price (USD)
1	1500	3	450,000
2	2000	3.5	520,000
3	1200	2.5	340,000
4	1800	3.2	500,000
5	1600	2.8	430,000
6	1400	2.2	480,000
7	1300	2.6	320,000
8	1700	3.3	480,000
9	2100	3.1	560,000
10	2300	3.4	600,000

3. In the **Descriptive Statistics** section, select **Input Range** as \$D\$2:\$D\$11 and **Output Range** as \$F\$3. Select the **Summary Statistics** checkmark. Select **OK**.

The screenshot shows the Microsoft Excel for the web interface. The ribbon is set to 'Home'. The spreadsheet contains the following data:

House	Area (sq. ft.)	Bank's Interest Rate (%)	Housing Price (USD)
1	1500	3	450,000
2	2000	3.5	520,000
3	1200	2.5	340,000
4	1800	3.2	500,000
5	1600	2.8	430,000
6	1400	2.2	480,000
7	1300	2.6	320,000
8	1700	3.3	480,000
9	2100	3.1	560,000
10	2300	3.4	600,000

The XLMiner Analysis ToolPak is open on the right side of the screen. The 'Descriptive Statistics' tab is selected. The configuration is as follows:

- Input Range: **\$D\$2:\$D\$11**
- Grouped By: ☒ Columns ☐ Rows
- ☒ Labels in First Row
- Output Range: **\$F\$3**
- ☒ Summary Statistics
- ☐ Confidence Level for Mean: 95 %
- ☐ Kth Largest: 1
- ☐ Kth Smallest: 1

The 'OK' button is highlighted with a red box.

4. All the descriptive statistics for the given data set will appear in Excel for the web. You can see that the computed mean for the given score is 470,000, the standard error is 31,091.26, and the median and mode are 480,000, respectively. The standard deviation is 93,273.79, the sample variance is 8,700,000,000, the kurtosis is (0.49), the skewness is (0.51), the range is 280,000, the minimum score is 320,000, the maximum score is 600,000, the sum is 42,30,0000, and the count is 9.

The screenshot shows a Microsoft Excel spreadsheet with a dataset of house prices. The data is organized in columns: House (A), Area (sq. ft.) (B), Bank's Interest Rate (%) (C), and Housing Price (USD) (D). The rows represent individual houses, numbered 1 to 10. The formula bar shows the value -0.492101240020383 in cell G8.

The XLMiner Analysis ToolPak is open, displaying the Descriptive Statistics output. The input range is \$D\$2:\$D\$11, and the output range is \$F\$3. The output table is highlighted with a red border.

	A	B	C	D	E	F	G	H
1	House	Area (sq. ft.)	Bank's Interest Rate (%)	Housing Price (USD)				
2	1	1500	3	450,000		Mean	470,000.00	
3	2	2000	3.5	520,000		Standard Error	31,091.26	
4	3	1200	2.5	340,000		Median	480,000.00	
5	4	1800	3.2	500,000		Mode	480,000.00	
6	5	1600	2.8	430,000		Standard Deviation	93,273.79	
7	6	1400	2.2	480,000		Sample Variance	8,700,000,000.00	
8	7	1300	2.6	320,000		Kurtosis	(0.49)	
9	8	1700	3.3	480,000		Skewness	(0.51)	
10	9	2100	3.1	560,000		Range	280,000.00	
11	10	2300	3.4	600,000		Minimum	320,000.00	
12						Maximum	600,000.00	
13						Sum	4,230,000.00	
14						Count	9.00	

The XLMiner Analysis ToolPak Descriptive Statistics output table is as follows:

	F	G
Mean		470,000.00
Standard Error		31,091.26
Median		480,000.00
Mode		480,000.00
Standard Deviation		93,273.79
Sample Variance		8,700,000,000.00
Kurtosis		(0.49)
Skewness		(0.51)
Range		280,000.00
Minimum		320,000.00
Maximum		600,000.00
Sum		4,230,000.00
Count		9.00

Congratulations! You've completed the lab Descriptive Statistics. Now, you are ready to move to the next topic.

Author(s)

- Dr. Rajendra Patil

Other Contributors

- Bhavika Chhatbar
- Rashi Kapoor



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